



Software Process Improvement –SEPI-471



PROCESS MODELS FOR SPI – Six Sigma Methodologies

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Agenda

- Process Models for SPI
 - Six Sigma
 - Lean Six Sigma
 - Six Sigma Methodologies
 - DMAIC
 - DMADV

Six Sigma (6σ)

Six Sigma is a set of methodologies and tools used to **improve business processes by reducing defects and errors, minimizing variation**, and increasing quality and efficiency.

Its approach is to identify and **remove the causes of defects in processes by minimizing process variability.**

Lean Six Sigma (6σ)

Lean Six Sigma is a methodology that combines two powerful process improvement techniques **Lean and Six Sigma**.

Lean focuses on minimizing waste and maximizing efficiency by identifying and eliminating non-value-adding activities.

Six Sigma (6σ)

It uses quality management techniques and tools such as the

-5 Whys

-Business process mapping

-DMAIC and DMADV methodologies

- Define, Measure, Analyze, Improve and Control -DMAIC
- Define, Measure, Analyze, Design, and Verify.
- DMAIC reduces/eliminates defects (reactive); DMADV prevents defects (proactive)

DMAIC

DMAIC (pronounced də-MAY-ick) is a data-driven technique used to improve processes.

It is a key tool of the **Six Sigma methodology**, but it can be implemented as a standalone improvement method or as part of other process initiatives like **Lean**.

DMAIC

“Lean” production originates from the Toyota Production System introduced in Japan by engineer Taiichi Ohno in the 1950s through 1980s.

The goal of **Lean** is to improve efficiency of a process and deliver the best product by **eliminating waste**.

Six Sigma is a Western concept that originated from **statistical modeling** of manufacturing processes themselves.



DMAIC

Lean Six Sigma” combines the principles of both Lean and Six Sigma with the overarching goal to reduce both waste and variation within a system using data and continuous quality improvement.

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DMAIC

This approach identify and eliminate causes of variation and improve processes

D - Define

M-Measure

A -Analyze

I – Improve

C- Control

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DMAIC

Step 1: Define

Objective: Clearly define the problem, project goals, scope, and stakeholders' requirements.

Activities:

Identify the problem or opportunity for improvement.

Define project goals and objectives.

Develop a project charter outlining the scope, resources, and timeline.

Identify key stakeholders and their requirements.

DMAIC

Step 2: Measure

Objective: Establish baseline performance metrics and collect data to quantify the current state of the process.

Activities:

Determine critical process inputs (variables) and outputs (metrics).

Develop data collection plan and measurement system.

Collect relevant data.

Analyze data to understand process performance and variability.

DMAIC

Step 3: Analyze

Objective: Identify root causes of problems or variations in the process.

Activities:

Use statistical analysis tools to identify patterns and trends in data.

Conduct root cause analysis (e.g., Fishbone diagram, 5 Whys) to determine underlying causes.

Prioritize root causes based on their impact on the process.

DMAIC

Step 4: Improve

Objective: Develop and implement solutions to address identified root causes and improve process performance.

Activities: Generate potential solutions through brainstorming and analysis. Select the best solution(s) based on effectiveness, feasibility, and sustainability. Develop an implementation plan with specific action steps, responsibilities, and timelines. Implement the chosen solution(s) and monitor their effectiveness.

DMAIC

Step 5: Control



The control phase is crucial to achieving sustainable change and requires tracking process performance.

Monitor Variations

Develop control plans to **ensure that the process remains stable** and within desired performance parameters.

Establish procedures for responding to out-of-control situations.

Document lessons learned and best practices for future reference.

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DMADV

It is a structured approach used in Six Sigma for creating new processes or products that meet customer requirements with minimal defects.

DMADV is often employed when existing processes need significant improvement or when developing entirely new processes

D - Define

M-Measure

A -Analyze

D– Design

V- Verify

DMADV

Define Phase:

Objective: Clearly define the project goals, scope, and customer requirements.

Activities:

Identify the project objectives and desired outcomes.

Define customer requirements and expectations.

Develop a project charter outlining the scope, resources, and timeline.

Identify key stakeholders and their inputs.

DMADV

Measure

Objective: Gather data and establish baseline performance metrics related to the project goals.

Activities:

Determine critical-to-quality characteristics (CTQs) based on customer requirements.

Develop measurement systems to quantify CTQs.

Collect relevant data to assess current performance.

Analyze data to identify gaps between current performance and desired outcomes.

DMADV

Analyze

Objective: Analyze data to understand root causes of potential issues or gaps identified in the Measure phase.

Activities:

Use statistical analysis tools to identify patterns and trends in data.

Conduct root cause analysis to determine underlying reasons for performance gaps.

Prioritize critical factors influencing process or product performance.

DMADV

Design

Objective: Develop and design new processes or products based on the insights gained from the Analyze phase.

Activities:

Generate innovative solutions to address identified root causes and meet customer requirements.

Design detailed process or product specifications.

Conduct simulations or prototypes to validate design concepts.

Develop an implementation plan for deploying the new process or product.

DMADV

Validate/Verify

Objective: Validate the effectiveness and reliability of the newly designed process or product.

Activities:

Conduct pilot testing or trial runs to verify the performance of the new process

Collect data to assess the conformance to specifications and customer requirements.

Analyze feedback from stakeholders and make necessary adjustments.

Establish control measures to ensure ongoing performance and quality.

DMADV VS DMAIC

Objective:

DMAIC aims to improve existing processes by identifying and eliminating defects or inefficiencies.

DMADV aims to design new processes, products, or services that meet or exceed customer requirements from the outset.

Focus:

DMAIC focuses on process optimization and improvement within existing processes.

DMADV focuses on creating new processes or products with high levels of quality and reliability.

DMADV VS DMAIC

Application:

DMAIC is applied when organizations have existing processes that need improvement.

DMADV is applied when organizations are developing new processes

Sequence of Phases:

In DMAIC, the phases are Define, Measure, Analyze, Improve, and Control.

In DMADV, the phases are Define, Measure, Analyze, Design, and Verify.

Lean Six sigma

Lean Six Sigma?

Lean Six Sigma is a methodology that combines two powerful process improvement techniques: Lean and Six Sigma.

Lean focuses on minimizing waste and maximizing efficiency by identifying and eliminating non-value-adding activities. This involves streamlining processes, reducing defects, improving quality, and optimizing resources to deliver more value with less effort.

On the other hand, Six Sigma is a statistical approach to process improvement that aims to reduce variation and defects by using data-driven decision making. It involves defining, measuring, analyzing, improving, and controlling processes to achieve consistent and predictable results.

Lean Six sigma

By combining the strengths of these two methodologies, Lean Six Sigma provides a comprehensive approach to process improvement that can be applied to any industry or sector. It is widely used in manufacturing, healthcare, finance, and service industries to improve efficiency, reduce costs, and enhance customer satisfaction.



Thank you

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